Static and kinetic friction

antocciphysics | TEACHER KEY | CPA

Learning target: Distinguish adjustable static friction from sliding friction.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 1, Day 14; 3 course days later.*
Ground pushes the runner’s foot forward with 50 N. This external force can accelerate the runner; the partner forces act on different objects.

02. Explain a worked example (worked)
Balanced 3 N pair, then balanced 5 N pair. Friction adjusts while the box remains at rest.

03. Below the limit (model)
Actual 8 N left; net 0 N, not 15 N friction.

04. Crossing the limit (practice)
18>15 N starts slipping; net 18-12=6 N right.

05. Can friction help? (reason)
Forward. It opposes relative slipping or tendency at the contact, not always the person’s motion.

06. Choose a method (mixed) *Review: Unit 1, Day 5; 12 course days later.*
Central position 22 m; interval 20-24 m. Assumes steady velocity and the stated start.

07. Exit check (exit)
First 12 N opposite push; second 9 N in the direction of the push.

08. STRETCH: Apply it another way (stretch)
4 N: rest possible with 4 N friction. 8 N: limiting equilibrium. 10 N: starts sliding; net 4 N in push direction once moving.

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01. Bring back an earlier idea (retrieve) *Review: Unit 1, Day 14; 3 course days later.*
Equal opposite 15 N forces. Craft acceleration 5 m/s²; asteroid 0.05 m/s² opposite. Compares this interaction's contribution, or assumes other forces negligible.

02. Explain a worked example (worked)
Balanced 3 N pair, then balanced 5 N pair. Friction adjusts while the box remains at rest.

03. Actual versus maximum (model)
N 49 N; max 19.6 N; actual 12 N opposite push; a 0.

04. Sliding case (practice)
f_k=.30×49=14.7 N; net 10.3 N; a 2.06 m/s².

05. Estimate from data (reason)
Means 8 N and 6 N;μ_s≈.40 andμ_k≈.30; static coefficient gives limiting threshold.

06. Choose a method (mixed) *Review: Unit 1, Day 5; 12 course days later.*
Central 44 m; half-width 1.3 m; interval [42.7, 45.3] m; full width 2.6 m. Constant velocity and bounded input ranges.

07. Exit check (exit)
N 98 N; static limit 49 N.30 N case: friction 30 N, a 0.60 N case: friction 39.2 N, a 2.08 m/s².

08. STRETCH: Apply it another way (stretch)
N=50 N, f_s, max=20 N; 22 exceeds it. Kinetic friction 15 N, acceleration 7/5=1.4 m/s². Steady sliding requires 15 N.